

EEGLAB / ERPLAB · SESSION 01



Assignment 01 | Getting Started with EEG Signals

Signal Processing Department · Interdisciplinary Schools

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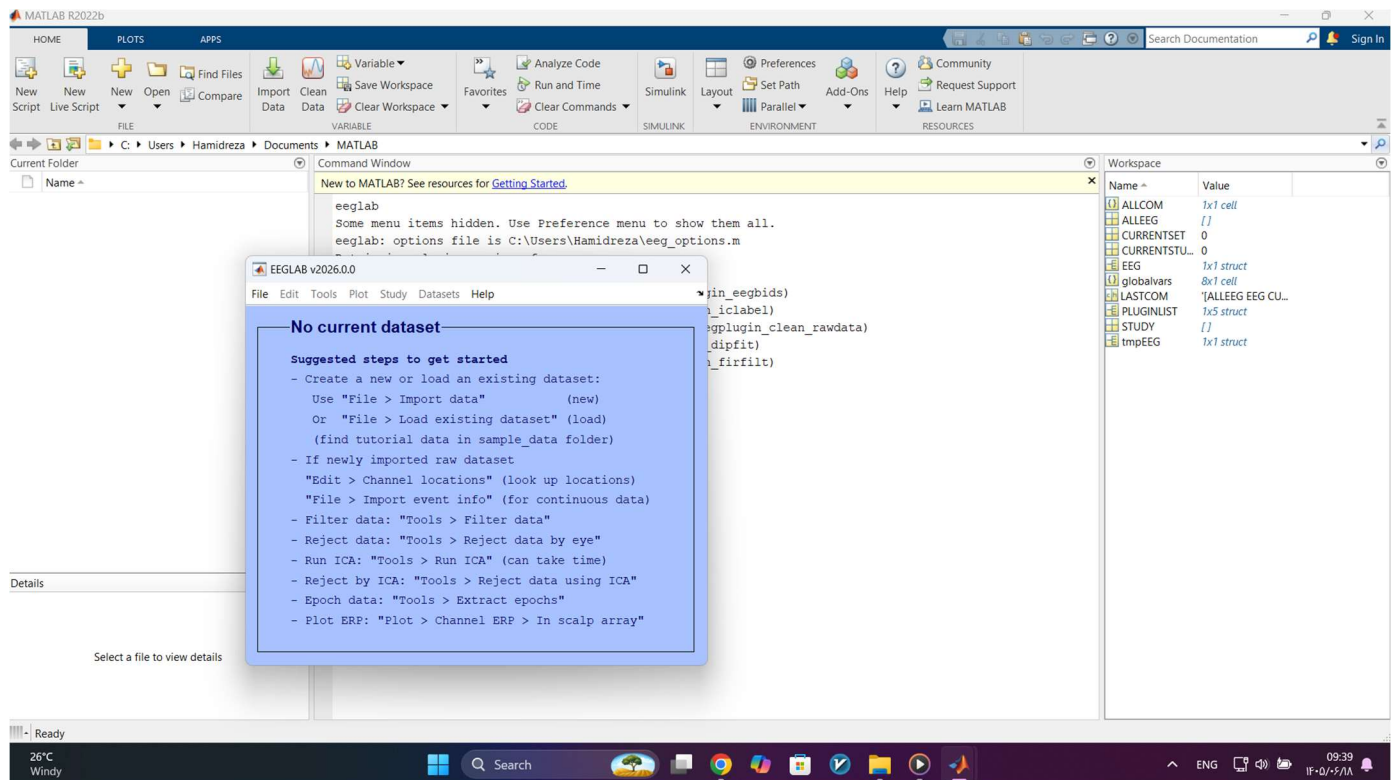
OBJECTIVE

Install the required software and connect the concepts from Session 01 to EEG frequency-band research.

01 MATLAB Installation

- Install MATLAB and launch it successfully on your computer.
- Submit: one screenshot showing MATLAB is installed and running.

Screenshot:



02 EEGLAB Installation

- Install EEGLAB in MATLAB and open the EEGLAB main interface.
- Submit: one screenshot showing the EEGLAB main window after launching it.

Screenshot: Above

03 EEG Frequency Bands · Mini Research Task

Choose one condition or disorder (e.g., ADHD, epilepsy, depression, or Alzheimer’s disease) and find one scientific article reporting changes in EEG frequency-band activity associated with that condition.

1 Alzheimer’s disease (AD) (with Mild Cognitive Impairment, MCI, also examined).	Condition / Disorder What condition was investigated?
2 analyses on Delta, Theta, and Alpha	EEG Frequency Band Which band(s) were reported? (Delta, Theta, Alpha, Beta, Gamma)
3 Increased spectral power (and coherence) in Delta and Theta bands. Decreased spectral power (and coherence) in the Alpha band. the Theta/Alpha ratio was the strongest discriminator between AD and controls.	Main Finding Was the reported activity increased, decreased, or otherwise altered?
4 Power spectral density and coherence analysis of Alzheimer’s EEG https://doi.org/10.1007/s11571-014-9325-x	Reference Provide the article title and DOI or link.

Important: Report the finding from the selected article. Do not treat a frequency-band pattern as a diagnostic rule or as universal for all individuals with that condition.

Submission: Submit this completed Word file with the two screenshots and your Task 03 answers. Include the article title and DOI/link.

SUBMISSION CHECKLIST

☐ MATLAB screenshot ☐ EEGLAB screenshot ☐ Task 03 answers ☐ Article title + DOI/link